

Detailed YFB Update Derivations

OpenAI Codex

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Table of contents

1 Purpose	2
2 Model Setup	2
3 Cox Taylor Expansion	3
4 Derivation of $q(\tilde{\beta}_k)$	3
4.1 Step 1: isolate factor k	3
4.2 Step 2: plug into the quadratic Cox surrogate	4
4.3 Step 3: read off the EBNM quadratic coefficients	5
4.4 Plain-language interpretation	5
4.5 Why this differs from LB	5
4.6 Code mapping	6
5 Derivation of $q(L_k)$	6
5.1 Step 1: identify which ELBO terms contain L_k	6
5.2 Step 2: isolate factor k in the reconstruction	6
5.3 Step 3: read off the EBNM coefficients	7
5.4 Plain-language interpretation	8
5.5 Why this differs from LB	8
5.6 Code mapping	8
6 Derivation of $q(F_k)$ — Mathematical YFB Version	8
6.1 Step 1: identify which ELBO terms contain F_k	8
6.2 Step 2: genomics contribution	9
6.3 Step 3: survival contribution	9
6.4 Step 4: combine the two information sources	10
6.5 Plain-language interpretation	10
7 Derivation of $q(F_k)$ — Current Implemented Version	11
7.1 Why this was done	11
7.2 Code mapping	11
8 Why the Three Updates Differ So Much	11
8.1 $q(\tilde{\beta}_k)$	11
8.2 $q(L_k)$	12

8.3	$q(F_k)$	12
9	Final Summary	12

1 Purpose

This note works through the YFB update derivations at the algebra level, starting from the ELBO and ending at the EBNM inputs used by the implementation.

The main goal is to make it clear why the three YFB updates are now structurally different:

- $q(\tilde{\beta}_k)$ changes because the survival covariate is now the observed projection score $(YF)_{ik}$
- $q(L_k)$ changes because L is no longer in the survival linear predictor
- $q(F_k)$ changes because F now affects both reconstruction and prognosis

Throughout, I will distinguish between:

- the **mathematical YFB derivation**
- the **current implemented YFB algorithm**

That distinction matters most for $q(F_k)$.

2 Model Setup

The genomics model remains

$$Y_{ij} \mid L, F, \tau \sim N \left(\sum_{k=1}^K l_{ik} f_{jk}, \tau_j^{-1} \right),$$

or in matrix form,

$$Y \approx LF^\top.$$

The YFB survival model replaces the LB predictor L **beta** with a projection-based predictor:

$$\eta_i = \sum_{k=1}^K (z_F)_{ik} \tilde{\beta}_k, \quad Z_F = YF, \quad (z_F)_{ik} = \sum_{j=1}^p y_{ij} f_{jk}.$$

So the Cox model becomes

$$h(t_i \mid y_i, F, \tilde{\beta}) = h_0(t_i) \exp(\eta_i).$$

Under mean-field variational inference we still write

$$q(L, F, \tilde{\beta}, \tau) = q(L) q(F) q(\tilde{\beta}) q(\tau).$$

The ELBO is

$$\mathcal{L} = E_q[\log p(Y \mid L, F, \tau)] + E_q[\log p(t, \delta \mid Y, F, \tilde{\beta})] - \text{KL}\{q(L) \parallel p(L)\} - \text{KL}\{q(F) \parallel p(F)\} - \text{KL}\{q(\tilde{\beta}) \parallel p(\tilde{\beta})\} - \text{KL}\{q(\tau) \parallel p(\tau)\}$$

The key structural fact is:

- the survival term depends on **F** and **tilde beta**
- the survival term does **not** depend on **L**

That one fact drives almost all of the downstream simplification.

3 Cox Taylor Expansion

As in the LB model, the Cox partial likelihood is handled by a second-order Taylor expansion around the current linear predictor.

Let:

- **u_i** be the Cox score
- **w_i** be the negative diagonal Hessian term
- **z_i** = **eta_i** + **u_i** / **w_i** be the working response

Then, up to constants,

$$\log p(t, \delta \mid \eta) \approx -\frac{1}{2} \sum_i w_i (z_i - \eta_i)^2.$$

This is the form that makes each factor update look like a weighted quadratic regression problem.

In YFB, the predictor is

$$\eta_i = \sum_{k=1}^K (z_F)_{ik} \tilde{\beta}_k.$$

So every YFB survival update is ultimately derived by isolating one factor-specific contribution inside this quadratic expression.

4 Derivation of $q(\tilde{\beta}_k)$

This is the cleanest YFB update, because the survival covariate is now explicit.

4.1 Step 1: isolate factor **k**

Write the YFB linear predictor as

$$\eta_i = (z_F)_{ik} \tilde{\beta}_k + \tilde{\eta}_i^{-k},$$

where $\tilde{\eta}^{-k}$ collects the contribution of all factors except **k**.

Equivalently, using the working-response notation used in the code,

$$z_i^{-k} = z_i - \sum_{k' \neq k} (z_F)_{ik'} E[\tilde{\beta}_{k'}].$$

In plain language: z_i^{-k} is the part of the working response that factor k still needs to explain.

4.2 Step 2: plug into the quadratic Cox surrogate

The relevant part of the survival objective is

$$-\frac{1}{2} \sum_i w_i (z_i - \eta_i)^2.$$

Substitute

$$\eta_i = (z_F)_{ik} \tilde{\beta}_k + \eta_i^{-k}.$$

Then

$$z_i - \eta_i = z_i - \eta_i^{-k} - (z_F)_{ik} \tilde{\beta}_k = z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k.$$

So the factor- k part becomes

$$-\frac{1}{2} \sum_i w_i \left(z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k \right)^2.$$

Expand the square:

$$\left(z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k \right)^2 = (z_i^{-k})^2 - 2z_i^{-k} (z_F)_{ik} \tilde{\beta}_k + (z_F)_{ik}^2 \tilde{\beta}_k^2.$$

Therefore

$$-\frac{1}{2} \sum_i w_i (z_i^{-k})^2 + \sum_i w_i z_i^{-k} (z_F)_{ik} \tilde{\beta}_k - \frac{1}{2} \sum_i w_i (z_F)_{ik}^2 \tilde{\beta}_k^2.$$

The first term does not depend on $\tilde{\beta}_k$, so it is absorbed into the constant.

What remains is

$$-\frac{1}{2} \left(\sum_i w_i (z_F)_{ik}^2 \right) \tilde{\beta}_k^2 + \left(\sum_i w_i z_i^{-k} (z_F)_{ik} \right) \tilde{\beta}_k + \text{const.}$$

4.3 Step 3: read off the EBNM quadratic coefficients

This has the standard form

$$-\frac{1}{2}A_{\tilde{\beta},k}\tilde{\beta}_k^2 + B_{\tilde{\beta},k}\tilde{\beta}_k + \text{const},$$

so

$$A_{\tilde{\beta},k} = \alpha \sum_i w_i (z_F)_{ik}^2$$

$$B_{\tilde{\beta},k} = \alpha \sum_i w_i z_i^{-k} (z_F)_{ik}.$$

The pseudo-observation and pseudo-noise are therefore

$$x_{\tilde{\beta},k} = \frac{B_{\tilde{\beta},k}}{A_{\tilde{\beta},k}} = \frac{\sum_i w_i z_i^{-k} (z_F)_{ik}}{\sum_i w_i (z_F)_{ik}^2},$$

$$s_{\tilde{\beta},k} = \frac{1}{\sqrt{A_{\tilde{\beta},k}}} = \frac{1}{\sqrt{\alpha \sum_i w_i (z_F)_{ik}^2}}.$$

4.4 Plain-language interpretation

This is just a one-dimensional weighted regression of the Cox working response onto the factor-k projection score.

- the denominator A is the weighted size of the predictor
- the numerator B is the weighted alignment between the residual response and the predictor
- the ratio B/A is the least-squares coefficient
- EBNM then shrinks that coefficient toward zero using the chosen prior

4.5 Why this differs from LB

In LB, the survival covariate was latent l_{ik} , so the update needed

$$E[l_{ik}^2]$$

rather than just l_{ik}^2 .

That was an error-in-variables correction: uncertainty in L propagates into uncertainty in **beta**.

In YFB, the covariate is treated as the fixed observed score $(z_F)_{ik}$ at the current outer iteration. So the derivation uses

$$(z_F)_{ik}^2$$

directly.

This is why YFB $q(\text{beta_k})$ is conceptually simpler than LB $q(\text{beta_k})$.

4.6 Code mapping

The code reuses the generic beta update in [update_beta.R](#), but in YFB it passes:

- `EL_k <- ZF[,k]`
- `EL2_k <- ZF[,k]^2`

from [fit_cox_on_yf.R](#).

So mathematically the derivation is cleaner than the API, but the implementation is equivalent.

5 Derivation of $q(L_k)$

This one changes the most structurally from LB.

5.1 Step 1: identify which ELBO terms contain L_k

Under YFB,

$$\eta_i = (y_i^\top F) \tilde{\beta}$$

contains `F` and `tilde beta`, but it does **not** contain `L`.

Therefore:

$$\frac{\partial}{\partial q(L_k)} E_q[\log p(t, \delta \mid Y, F, \tilde{\beta})] = 0.$$

So the $q(L_k)$ update comes entirely from:

$$E_q[\log p(Y \mid L, F, \tau)] - \text{KL}\{q(L_k) \parallel p(L_k)\}.$$

This is the key simplification.

5.2 Step 2: isolate factor k in the reconstruction

Define the partial residual with factor k removed:

$$R_k = Y - \sum_{k' \neq k} E[l_{k'}] E[f_{k'}]^\top.$$

Then the reconstruction for factor k is

$$R_k \approx l_k f_k^\top.$$

At the element level,

$$r_{k,ij} \approx l_{ik} f_{jk}.$$

The genomics log-likelihood contribution is

$$-\frac{1}{2} \sum_{i,j} \tau_j E[(r_{k,ij} - l_{ik} f_{jk})^2].$$

Expand the square:

$$(r_{k,ij} - l_{ik} f_{jk})^2 = r_{k,ij}^2 - 2r_{k,ij} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2.$$

Take expectations under mean-field:

$$E[r_{k,ij} l_{ik} f_{jk}] \approx r_{k,ij} E[l_{ik}] E[f_{jk}]$$

and

$$E[l_{ik}^2 f_{jk}^2] = E[l_{ik}^2] E[f_{jk}^2].$$

So the factor- k contribution becomes

$$-\frac{1}{2} \sum_{i,j} \tau_j (r_{k,ij}^2 - 2r_{k,ij} E[l_{ik}] E[f_{jk}] + E[l_{ik}^2] E[f_{jk}^2]).$$

Now isolate the terms involving a particular l_{ik} :

$$-\frac{1}{2} \left(E[l_{ik}^2] \sum_j \tau_j E[f_{jk}^2] - 2E[l_{ik}] \sum_j \tau_j r_{k,ij} E[f_{jk}] \right) + \text{const.}$$

5.3 Step 3: read off the EBNM coefficients

This again has the standard quadratic form

$$-\frac{1}{2} A_{L,k} E[l_{ik}^2] + B_{L,k}(i) E[l_{ik}] + \text{const},$$

with

$$A_{L,k} = \sum_j \tau_j E[f_{jk}^2]$$

$$B_{L,k}(i) = \sum_j \tau_j r_{k,ij} E[f_{jk}].$$

Therefore

$$x_{L,ik} = \frac{B_{L,k}(i)}{A_{L,k}}, \quad s_{L,k} = \frac{1}{\sqrt{A_{L,k}}}.$$

5.4 Plain-language interpretation

This is ordinary weighted matrix-factorization algebra.

For each patient i , the update asks:

- given the current factor column $\mathbf{f}_k$
- and given the residual matrix after removing all other factors
- what value of $l_{\{ik\}}$ best explains the remaining reconstruction error?

The answer is turned into an EBNM pseudo-observation and then shrunk according to the prior on L .

5.5 Why this differs from LB

In LB, L_k was supervised. Its update had:

- a genomics contribution
- a survival contribution involving w_i , z_i^{-k} , $E[\beta_k]$, and $E[\beta_k^2]$

In YFB, all of those disappear because L is no longer inside the Cox predictor.

So YFB $q(L_k)$ is much simpler than LB $q(L_k)$.

5.6 Code mapping

This derivation is implemented directly in [update_L_surv_YFB.R](#).

The code computes:

- `A_gen <- sum(Tau * EF2_k)`
- `B_L <- R_k %*% (Tau * EF_k)`

which are exactly the expressions above.

6 Derivation of $q(F_k)$ — Mathematical YFB Version

This is the most important derivation because F_k is the bridge between reconstruction and prognosis.

6.1 Step 1: identify which ELBO terms contain F_k

Unlike L_k , the factor column F_k appears in both:

1. the genomics reconstruction $Y \approx LF^{\text{top}}$
2. the survival predictor $\eta = YF^{\text{tilde}}\beta$

So the $q(F_k)$ update is dual-source in the mathematical YFB model.

6.2 Step 2: genomics contribution

Using the same residual definition $R_{\mathbf{k}}$, the genomics term is

$$-\frac{1}{2} \sum_{i,j} \tau_j E[(r_{k,ij} - l_{ik} f_{jk})^2].$$

Expanding and isolating terms involving $\mathbf{f}_{\{\mathbf{j}\mathbf{k}\}}$ gives

$$-\frac{1}{2} \left(E[f_{jk}^2] \tau_j \sum_i E[l_{ik}^2] - 2E[f_{jk}] \tau_j \sum_i r_{k,ij} E[l_{ik}] \right) + \text{const.}$$

So the genomics-side coefficients are

$$A_{F,\text{gen},k}(j) = \tau_j \sum_i E[l_{ik}^2]$$

$$B_{F,\text{gen},k}(j) = \tau_j \sum_i r_{k,ij} E[l_{ik}].$$

6.3 Step 3: survival contribution

Now use the YFB predictor

$$\eta_i = \sum_{k'} (z_F)_{ik'} \tilde{\beta}_{k'}.$$

For factor $\mathbf{k}$,

$$(z_F)_{ik} = \sum_j y_{ij} f_{jk}.$$

So each $\mathbf{f}_{\{\mathbf{j}\mathbf{k}\}}$ enters linearly into the survival predictor.

Write

$$\eta_i = y_{ij} f_{jk} \tilde{\beta}_k + \eta_i^{-j,k},$$

where $\eta_i^{-j,k}$ collects all parts of the predictor that do not depend on the specific coefficient $\mathbf{f}_{\{\mathbf{j}\mathbf{k}\}}$.

The Cox quadratic surrogate contributes

$$-\frac{1}{2} \sum_i w_i \left(z_i^{-k} - y_{ij} f_{jk} \tilde{\beta}_k \right)^2.$$

Expand the square:

$$\left(z_i^{-k} - y_{ij}f_{jk}\tilde{\beta}_k\right)^2 = (z_i^{-k})^2 - 2z_i^{-k}y_{ij}f_{jk}\tilde{\beta}_k + y_{ij}^2f_{jk}^2\tilde{\beta}_k^2.$$

Taking expectations under mean-field gives

$$-\frac{1}{2} \left(E[f_{jk}^2]E[\tilde{\beta}_k^2] \sum_i w_i y_{ij}^2 - 2E[f_{jk}]E[\tilde{\beta}_k] \sum_i w_i z_i^{-k} y_{ij} \right) + \text{const.}$$

So the survival-side coefficients are

$$A_{F,\text{surv},k}(j) = E[\tilde{\beta}_k^2] \sum_i w_i y_{ij}^2$$

$$B_{F,\text{surv},k}(j) = E[\tilde{\beta}_k] \sum_i w_i z_i^{-k} y_{ij}.$$

6.4 Step 4: combine the two information sources

With mixing weight `alpha`,

$$A_{F,k}(j) = (1 - \alpha)A_{F,\text{gen},k}(j) + \alpha A_{F,\text{surv},k}(j)$$

$$B_{F,k}(j) = (1 - \alpha)B_{F,\text{gen},k}(j) + \alpha B_{F,\text{surv},k}(j).$$

Then the EBNM inputs are

$$x_{F,jk} = \frac{B_{F,k}(j)}{A_{F,k}(j)}, \quad s_{F,jk} = \frac{1}{\sqrt{A_{F,k}(j)}}.$$

6.5 Plain-language interpretation

This update asks:

- which genes should define factor `k` so that it still reconstructs the data well,
- while also making the projection score `Y f_k` prognostic?

That is the clean theoretical heart of YFB.

This is the derivation that makes YFB different from simply “fit an unsupervised factorization and then do Cox regression afterward.”

7 Derivation of $q(F_k)$ — Current Implemented Version

The current implementation does **not** use the full dual-source F update.

Instead, it sets

$$\alpha_F = 0.$$

So the actual update used in fitting is

$$A_{F,k}(j) = A_{F,\text{gen},k}(j) = \tau_j \sum_i E[l_{ik}^2]$$

$$B_{F,k}(j) = B_{F,\text{gen},k}(j) = \tau_j \sum_i r_{k,ij} E[l_{ik}].$$

This means the implemented $q(F_k)$ is genomics-only, even though the mathematical YFB derivation says it should be dual-source.

7.1 Why this was done

The survival contribution to F created a positive-feedback instability:

- F determines $ZF = YF$
- ZF determines beta
- beta then pushes back on F

Empirically, this was unstable, so the current implementation keeps the YFB survival parameterization but stabilizes the factor update by learning F from genomics alone.

7.2 Code mapping

This is exactly what `update_F_surv_YFB.R` implements.

The function computes both:

- `A_gen`, `B_gen`
- `A_surv`, `B_surv`

but `fit_cox_on_yf.R` calls it with `alpha = 0`, so only the genomics terms are active.

8 Why the Three Updates Differ So Much

It is worth stating this plainly, because it is the main conceptual change from LB.

8.1 $q(\tilde{\text{beta}}_k)$

The survival coefficient is now regressed on the observed projection score ZF_k . That makes the update more direct than in LB.

8.2 $q(L_k)$

L no longer enters the survival predictor at all.
That makes the update simpler than in LB.

8.3 $q(F_k)$

F now determines both:

- reconstruction quality
- the prognostic score basis $ZF = YF$

So F becomes the central object in the YFB derivation.

9 Final Summary

The detailed YFB derivational story is:

1. Replace the LB survival predictor $L\beta$ with the projection-based predictor $Y\tilde{\beta}$.
2. That makes $q(\tilde{\beta}_k)$ a Cox-on-observed-score update rather than Cox-on-latent-score.
3. That removes survival entirely from $q(L_k)$.
4. That moves the dual-source supervision burden onto $q(F_k)$.
5. The current implementation keeps steps 1–3, but uses a stabilized genomics-only version of step 4.

So the mathematically intended YFB model and the current working implementation are close, but not identical.

That is the most important distinction to keep in mind when explaining the model.